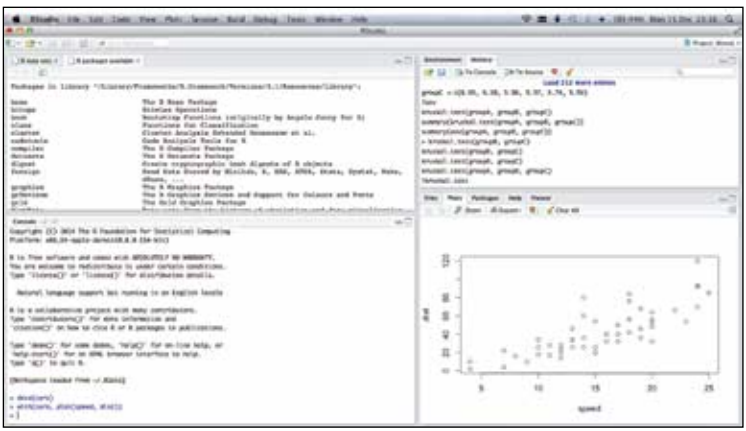
R is specialized for these uses because it offers packages that allow users to clean data, and run models or tests on data inputted into it. The models provide output that demonstrates how well the data fits the model and also useful graphs and charts to make sense of data. This type of specialised functionality makes R an ideal language to test statistical concepts, and goes hand in hand with learning statistics.

It is no wonder then that R has become a huge success in the data sciences, surpassing even Python in utility. Of course, in all fairness to Python, which is beautiful, or the more recent Julia, which lets you work with algorithms in ways that most other languages cannot, R has been around a lot longer. At the time of its release R was the only language written exclusively for statistical analysis, being an implementation of S.



R Studio has a better look and feel to it.

This gave it an edge over other languages and tools that entered the market later. As it was the first statistical tool on the scene, it gained a loyal user base that grew comfortable with it and wrote packages to continually improve it. It has had a head start of over 10 years, and has amassed a huge library geared towards statistical programming.

But this is only part of the story. Its constantly being updated means that R has kept up with the latest statistical techniques, and in addition to basic statistical models, is also the go to program for Machine Learning techniques - pattern recognition, neural network algorithms, clustering, random forests and deep learning.